TASK 4 — STEP-BY-STEP GUIDE

DESIGN AN INTERACTIVE iOS PROTOTYPE WITH FIGMA

Course: Geo Programming | Week 5 | Suggested time: 45–60 minutes | Individual work

1. Learning Outcomes

Identify justified screens; build a clickable Figma prototype; connect all necessary navigation paths; audit every use case and functional requirement against visible UI actions and feedback; plan SwiftUI View/controller responsibilities.

2. Scenario & Inputs

School Management System (2026–2027): staff browse students, subjects, teachers and enrollments; add/edit students; assign subjects; enroll students; enter/update marks from 0 to 10; and inspect results. Duplicate enrollments must be rejected.

Use the functional requirements (Task 1), use cases (Task 2) and OOP models (Task 3). Design and prototype first; do not implement the full SwiftUI app in this task.

Step 1 — Identify Screens and Their Purposes

Derive screens from user goals, not one screen per class. Three worked examples:

Screen ID / name	Purpose	Related use case
S01 Student List	Browse student records	UC: Browse Students
S02 Student Detail	Inspect student, subjects and marks	UC: Inspect Student Results
S03 Student Form	Add or edit student	UC: Add/Edit Student

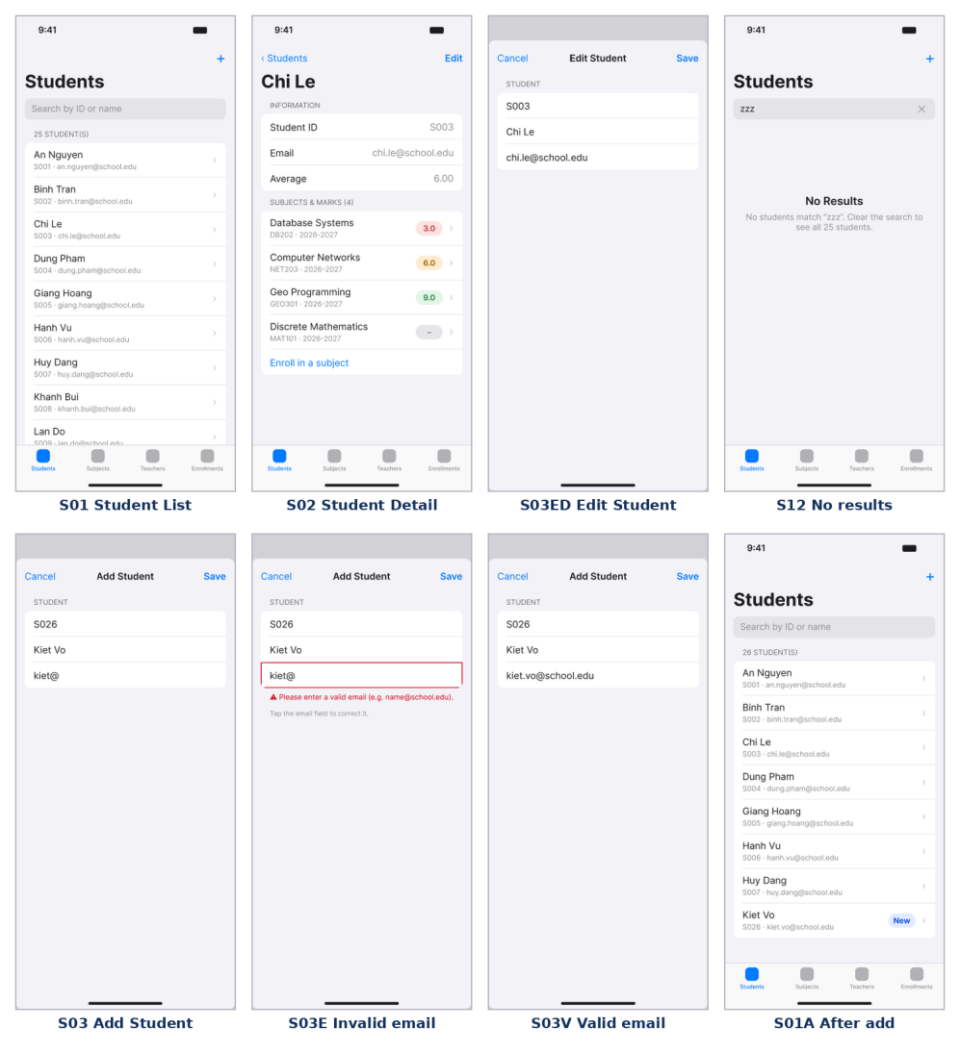
YOUR TURN — Identify ALL remaining justified screens

List every screen needed for remaining use cases (subjects, teachers, enrollment, marks, results), including error/empty states when distinct designs are necessary. Add rows as needed.

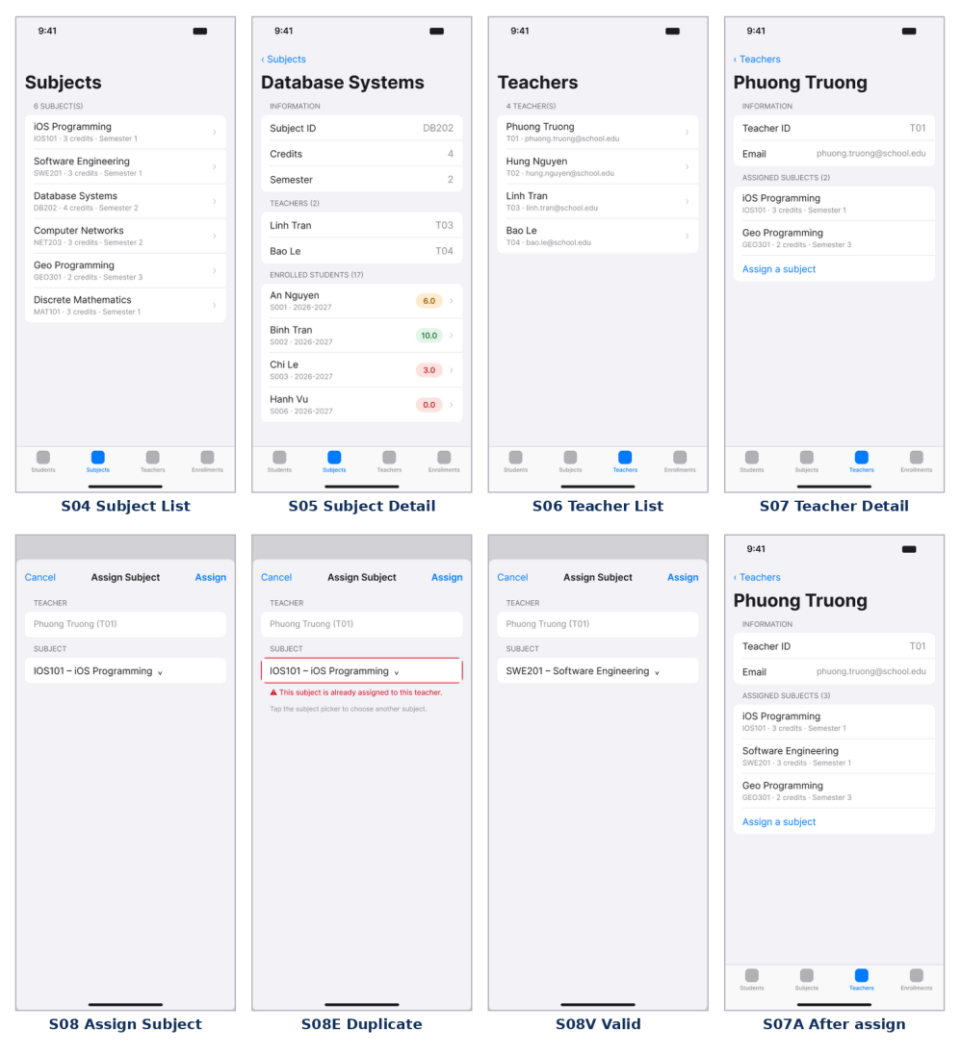
Screen ID / name	Purpose	UC ID(s)
S04 Subject List	Browse subjects (ID, title, credits, semester)	UC-09
S05 Subject Detail	Show a subject's teachers and enrolled students with marks	UC-09
S06 Teacher List	Browse teachers and their number of subjects	UC-09
S07 Teacher Detail	Show assigned subjects; remove an assignment	UC-07, UC-09
S08 Assign Subject (overlay)	Pick a subject for a teacher; duplicate-assignment error state	UC-07
S09 Enrollment List	Browse and search all enrollments	UC-02, UC-03
S10 Enrollment Form	Choose student, subject and year; duplicate-enrollment error state	UC-02
S11 Update Mark	Enter or update a mark; invalid-mark error state	UC-03
S12 Empty / no-result states	“No matching students” with a clear (✕) action	UC-01, UC-04
S03ED Edit Student	Edit an existing student (pre-filled form)	UC-06

S01A, S02U, S02E, S07A result states	Show the saved result after Add, Save Mark, Enroll and Assign	UC-05, UC-03, UC-02, UC-07
S03E, S08E, S10E, S11E error states	Invalid email, duplicate assignment, duplicate enrollment, mark outside 0–10	UC-05, UC-07, UC-02, UC-03

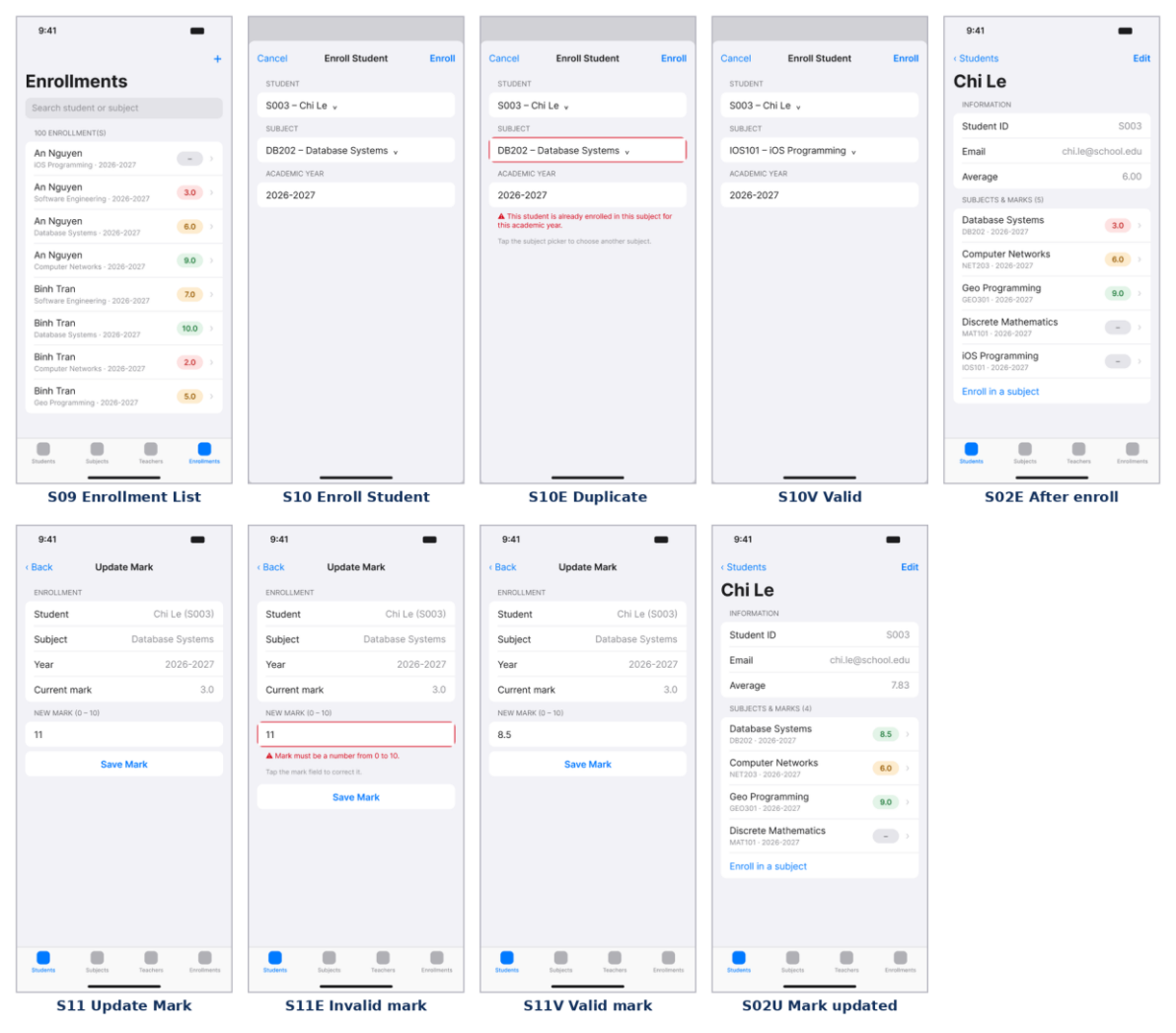
Figma screens – Students (S01, S02, S03ED, S12, S03 → S03E → S03V → S01A):



Figma screens – Subjects and Teachers (S04–S07, S08 → S08E → S08V → S07A):



Figma screens – Enrollments and marks (S09, S10 → S10E → S10V → S02E, S11 → S11E → S11V → S02U):



Steps 2–4 — Design Screens and Connect Them in Figma

These three former paper-sketch steps are replaced by ONE practical Figma workflow. Use https://www.figma.com/ and create a design file named GeoProgramming_Task4_StudentID.

A. Create the design file and screen frames

1. Create a new Figma Design file. Add a page named Screens.
2. Press F (Frame) and select a consistent iPhone-sized preset. Name each frame with its screen ID and purpose.
3. Use shapes, text and reusable components to create the screen layout. Include data labels, input fields, actions and user feedback.
4. Create all justified screens identified in Step 1, not just the three examples.

Three guided screen examples

Frame	Visible elements	Actions / feedback
S01 Student List	Title, search field, student rows, Add (+)	Tap row; tap Add; empty-list message
S02 Student Detail	Student ID/name, enrolled subjects, marks, Edit	Tap Edit; navigate back
S03 Student Form	ID, name, email, Save, Cancel	Save success; required-field error

B. Connect screens with Figma Prototype

1. Switch to the Prototype tab. Select an interactive element, then drag its prototype connector to the destination frame.
2. Set trigger On click / On tap and action Navigate to (or Open overlay for a modal).
3. Define a starting frame. Add Back, Cancel and Save success paths.
4. Use separate frames or overlays to demonstrate invalid input, duplicate enrollment and other feedback.
5. Click Present / Play and test the prototype end to end.

Three guided prototype connections

From	Tap / trigger	To / outcome
S01 Student List	Tap student row	S02 Student Detail
S01 Student List	Tap Add (+)	S03 Student Form
S02 Student Detail	Tap Edit	S03 Student Form (edit state)

YOUR TURN — Build and connect ALL remaining screens

Create the rest of the screens and interactions in Figma. Test forward and return routes; include success, cancellation, empty data and validation feedback. Record unresolved links below.

Source frame	Element / action	Destination / state	Tested?
S02 Student Detail	Tap “Enroll in a subject”	S10 Enrollment Form	Yes – Present mode
S02 Student Detail	Tap an enrolled subject	S11 Update Mark	Yes – Present mode
S02 Student Detail	Tap Edit	S03ED Edit Student (Save / Cancel → S02)	Yes – Present mode
S03 Student Form	Tap Save with email “kiet@”	S03E invalid-email error	Yes – Present mode
S03E → S03V	Tap the email field, then Save	S01A list with the new row Kiet Vo (26 students)	Yes – Present mode
S03 Student Form	Tap Cancel	S01 Student List unchanged	Yes – Present mode
S01 Student List	Tap the search field	S12 no-result state; ✕ → S01	Yes – Present mode
S04 Subject List	Tap a subject row	S05 Subject Detail	Yes – Present mode
S05 Subject Detail	Tap an enrolled student	S11 Update Mark	Yes – Present mode
S06 Teacher List	Tap a teacher row	S07 Teacher Detail	Yes – Present mode
S07 Teacher Detail	Tap “Assign a subject”	S08 Assign Subject (sheet)	Yes – Present mode
S08 Assign Subject	Tap Assign (IOS101 already assigned)	S08E duplicate error	Yes – Present mode
S08E → S08V	Pick SWE201, tap Assign	S07A teacher with 3 subjects	Yes – Present mode
S09 Enrollment List	Tap Add (+) / tap a row	S10 Enrollment Form / S11 Update Mark	Yes – Present mode
S10 Enrollment Form	Tap Enroll (DB202 again)	S10E duplicate-enrollment error	Yes – Present mode
S10E → S10V	Pick IOS101, tap Enroll	S02E with the new IOS101 row	Yes – Present mode
S11 Update Mark	Tap Save Mark with 11	S11E invalid-mark error, mark stays 3.0	Yes – Present mode
S11E → S11V	Enter 8.5, tap Save Mark	S02U with DB202 = 8.5, average 7.83	Yes – Present mode
Tab bar (all main screens)	Tap Students / Subjects / Teachers / Enrollments	S01 / S04 / S06 / S09	Yes – Present mode

Step 5 — Audit Use Cases, Functions and UI Mapping

Goal: prove that EVERY functional requirement (FR) from Task 1 and EVERY use case (UC) from Task 2 is represented in the Figma prototype. Do not treat a screen title alone as proof: identify the exact control, action and observable feedback.

A. Make a master inventory

1. Copy ALL FR IDs and short descriptions from Task 1.
2. Copy ALL UC IDs from Task 2, including alternate/error flows.
3. For each FR, identify the related UC(s). For each UC, identify the FR(s) it satisfies. Mark missing links as gaps, not as completed.

B. Trace each requirement to the interface

4. Identify the Figma frame ID, exact UI control, user action, destination/state and expected feedback.
5. Check the prototype in Present mode. Mark Pass only if the intended interaction and feedback are actually represented.
6. Record missing functionality, missing screens and unclear error states; revise Figma and recheck.

Three worked mapping examples (illustrative IDs; use your actual Task 1–2 IDs)

FR / UC	Figma frame + UI action	Success / error feedback	Status
Browse students / Browse Students	S01 → open Students list	Rows displayed / “No students yet”	Example
Edit student / Add/Edit Student	S03 → enter values → Save	Updated detail / invalid email message	Example
Update mark / Update Mark	Enrollment Detail → enter mark → Save	Updated mark / “Mark must be 0–10”	Example

YOUR TURN — Complete the FULL traceability matrix

Enter EVERY remaining FR and UC, including alternate flows. Extend the table until there are no uncovered requirements.

FR ID	UC ID / flow	Frame ID	UI control + action	Feedback / destination	Pass / gap
FR-01	UC-01	S01	Open the Students tab	Rows with ID and name	Pass (empty list: gap, see log)
FR-02	UC-04	S01 → S12	Tap the search field	No-result message; ✕ clears	Pass
FR-03	UC-05	S01 → S03 → S01A	Tap + → fill form → Save	New row Kiet Vo in S01A	Pass
FR-04	UC-06	S02 → S03ED	Tap Edit → Save	Back to S02	Pass
FR-05	UC-09	S04, S05, S06, S07	Open Subjects / Teachers tab → tap a row	Subject fields, its teachers and students	Pass
FR-06	UC-07	S07 → S08 → S07A	Assign a subject → pick → Assign	Subject added; duplicate message on S08E	Pass (Remove: gap, see log)
FR-07	UC-02	S02 / S09 → S10 → S02E	Choose student, subject, year → Enroll	New IOS101 row on S02E	Pass

FR-08	UC-02 (duplicate flow)	S10 → S10E	Enroll DB202 again	“Already enrolled...” error; no new row	Pass
FR-09	UC-03	S11V → S02U	Type 8.5 → Save Mark	Badge 8.5, average 7.83	Pass
FR-10	UC-03 (invalid mark)	S11 → S11E	Type 11 → Save Mark	“Mark must be a number from 0 to 10”; mark stays 3.0	Pass
FR-11	UC-08	S01 → S02	Tap a student	Subjects, marks and average	Pass
FR-12	UC-05 / UC-06 (invalid data)	S03 → S03E	Save with email “kiet@”	Email error; form stays open	Pass

Step 5 — Checklist for Functional Coverage (continued)

Use this checklist AFTER testing the Figma prototype. A requirement is covered only when the screen, action and feedback can be demonstrated.

☑ Every FR from Task 1 appears in the traceability matrix.

☑ Every UC from Task 2 appears, including alternate and error flows.

☑ Each mapped item names a specific Figma frame and interactive control.

☑ Every key action has a visible success result or destination.

☑ Invalid marks (outside 0–10) show a clear error and do not imply a successful save.

☑ Duplicate enrollment produces an error state and does not imply a duplicate record was created.

☑ Required fields and other documented validation rules have feedback.

☑ Empty data and cancel/back navigation are demonstrated where relevant.

☑ All Figma prototype connections have been tested using Present mode.

☑ Any unsupported requirement is marked GAP and has a revision plan.

Gap log — identify and fix ALL uncovered items

FR / UC	What is missing?	Figma change required	Retest result
FR-03 / UC-05	After Save the list looked the same, so the new student was not visible	Added S01A with the Kiet Vo row and “26 student(s)”; S03V Save → S01A	Pass
FR-04 / UC-06	Edit opened the Add Student form with S026	Added S03ED Edit Student pre-filled with Chi Le; Save / Cancel → S02	Pass
FR-09 / UC-03	After Save Mark S02 still showed 3.0	Added S02U with DB202 = 8.5 and average 7.83	Pass
FR-07 / UC-02	After Enroll no new subject appeared	Added S02E with the IOS101 row (5 subjects)	Pass
FR-06 / UC-07	After Assign the teacher still had 2 subjects	Added S07A with SWE201 (3 subjects)	Pass
FR-08, FR-10, FR-12	Error frames were a dead end	Tapping the red field on S03E / S08E / S10E / S11E opens the corrected frame S03V / S08V / S10V / S11V	Pass

FR-06 / UC-07 (Remove)	Swipe to remove an assigned subject is not prototyped	Plan: add a swipe-left “Remove” state on S07 rows (On drag)	GAP – done in SwiftUI with swipeActions
FR-01 / UC-01 (empty list)	“No students yet” cannot be reached with the sample data	Plan: copy S12 layout as an empty-list frame	GAP – only the no-result state (S12) is shown

Student reflection

Which FR or UC was hardest to map to an interface action? Explain your design decision in 3–5 sentences.

FR-10 / UC-03 (invalid mark) was the hardest to map, because a Figma prototype cannot really check what the user types. I made three frames for the same screen: S11 with the value 11, S11E with the red message “Mark must be a number from 0 to 10” while the current mark stays 3.0, and S11V with the corrected value 8.5. Save Mark on S11 goes to the error frame, tapping the red field opens S11V, and its Save Mark goes to S02U where the badge shows 8.5. This makes it clear that nothing is saved on error, and I reused the same pattern for the duplicate enrollment, duplicate assignment and invalid email cases.

Step 6 — Plan SwiftUI Views and Controller Responsibilities

Use the Figma design as a guide for later SwiftUI implementation. UI elements display data and collect actions; validation and business rules belong in appropriate model/service/controller logic.

Three worked examples

Proposed Swift file / component	Responsibility	Business rule / service
StudentListView.swift	Render student rows and navigation	StudentController.loadStudents()
StudentFormView.swift	Collect input; show errors	StudentController.saveStudent(...)
EnrollmentView.swift	Choose student/subject; show result	EnrollmentService.enroll(...) with duplicate check

YOUR TURN — Identify ALL remaining views and coordinating components

Figma frame	SwiftUI View file	Controller / service	Validation / state owner
S01 Student List	StudentListView.swift	SchoolController.searchStudents(_:)	SchoolController
S02 Student Detail	StudentDetailView.swift	enrollments(of:), averageMark(of:)	SchoolController
S03 Student Form	StudentFormView.swift	addStudent / updateStudent	Student + Validator
S04–S05 Subjects	SubjectListView.swift (SubjectDetailView)	subjects, teachers(of:), enrollments(of:)	SchoolController
S06–S08 Teachers	TeacherListView.swift (TeacherDetailView, AssignSubjectView)	assign / unassign	Teacher.assign
S09–S10 Enrollments	EnrollmentListView.swift, EnrollmentFormView.swift	enroll(studentID:subjectID:academicYear:)	SchoolController (duplicate) + Validator (year)
S11 Update Mark	MarkEditView.swift	updateMark(enrollmentID:markText:)	Enrollment.updateMark
App / tabs	SchoolManagementApp.swift, MainTabView.swift	One shared @Observable SchoolController	The app owns the state

Step 7 — Review, Test and Submit

Open the Figma prototype in Present mode and test every mapped FR/UC. Confirm screens, navigation, Save/Cancel paths and validation feedback. Fix all recorded gaps before submission.

Three worked review examples

Review item	Test	Expected result
Coverage	Open Student List from entry screen	Student rows / empty state visible
Navigation	Open Student Form, then Cancel	Return without saving
Validation	Try enrolling the same student twice	Duplicate warning state visible

YOUR TURN — Record ALL remaining checks

FR / UC / screen	Test action	Pass / gap	Revision
FR-10 / UC-03 / S11	Enter 11 → Save Mark	Pass – S11E error, mark stays 3.0	Added S11V → S02U for the corrected path
FR-08 / UC-02 / S10	Enroll Chi Le in DB202 2026-2027 again	Pass – S10E error, no new row	Added S10V → S02E
FR-12 / UC-05 / S03	Save with email “kiet@”	Pass – S03E email error	Empty name / existing ID use the same error pattern
FR-06 / UC-07 / S08	Assign IOS101 to Phuong Truong again	Pass – S08E duplicate error	Added S08V → S07A
UC-08 / S01 → S02	Open Chi Le, then tap ‹ Students	Pass – back to S01	None
FR-03 / UC-05 / S01A	Add Kiet Vo with a valid email	Pass – new row in S01A	Added S01A
FR-02 / UC-04 / S12	Tap search, then ✕	Pass – no-result state, then full list	None

Required Deliverables

☑ Shareable Figma design file link (view access enabled for instructor). Design: https://www.figma.com/design/cVPDvh0ojB1GAvPikuTDup/GeoProgramming_Task4_SESEIU22029

Prototype (Present mode, starts at S01): https://www.figma.com/proto/cVPDvh0ojB1GAvPikuTDup/GeoProgramming_Task4_SESEIU22029?node-id=3-2&starting-point-node-id=3-2

☑ Clickable Figma prototype with complete navigation and feedback states.

☑ Screen inventory with related UC IDs.

☑ Completed FR–UC–screen–action–feedback traceability matrix and checklist.

☑ Gap log and proposed SwiftUI View/controller structure.

Submit the Figma link and export your traceability worksheet as Task4_Figma_StudentID.pdf with Lab 01.